

Optimizing Columbia's Microtransit Shuttle Service with Batch Optimization

IEOR 4418 Project

Abstract

This report presents a dynamic batching and routing framework for Columbia's night shuttle service by integrating historical ride request data, road networks from OpenStreetMap, and a pickup-and-delivery Traveling Salesman Problem formulation. Our system performs rolling-window batching of ride requests, assignment of batches to vehicles, and Gurobi-based route optimization. Additionally, a vehicle supply plan is incorporated to reflect real fluctuations in fleet availability. We evaluate the model by using October 2025 data to conduct sensitivity analyses on varying rolling time windows and vehicle supply levels. We then discuss the results to demonstrate the effectiveness of our batch optimization solution and highlight opportunities for future enhancement.

Contents

1	Introduction	1
2	Problem Scope	1
3	Data and Preprocessing	1
4	Model Assumptions	2
5	Modeling Framework	3
5.1	Distance and Speed Modeling	3
5.2	Batching and Sub-Batching	3
5.3	Vehicle Assignment	3
5.4	Pickup-and-Delivery TSP Formulation	3
5.5	Successor-Based Route Reconstruction	3
5.6	Dispatch of Vehicles	4
6	Model Performance and Validation	4
6.1	Sensitivity Analysis	4
6.1.1	Test 1: Effect of Batching Window Size	4
6.1.2	Test 2: Effect of Vehicle Supply Levels	4
6.2	Confidence Interval Evaluation	5
6.3	Model Stability Under Temporal Demand Variation	6
7	Discussion	6
8	Conclusion	6
	References	7
	Appendix	8

1 Introduction

Columbia Transportation offers an on-demand micro transit shuttle service to provide students and staff a safer alternative to walking alone at night. Despite the availability of this free service, riders frequently experience long wait times or are simply rejected by the system due to seat unavailability during peak demand hours. The service’s struggle to meet demand diminishes the shuttle’s reliability and discourages riders from using the shuttle.

To better understand these challenges, we analyze operational data from Columbia’s ride request system by examining demand patterns and wait times against vehicle supply levels to understand the current system’s behavior. Using these insights, we explore data-driven strategies to reduce user wait times and therefore improve overall system performance.

2 Problem Scope

We first conducted a preliminary data analysis using the ride request data provided by Columbia Transportation and Via. From January to October of this year, observed wait times during door-to-door hours ranged from 0 minutes to 89 minutes with a median of 20 minutes and were slightly higher than wait times during service-to-service as shown in Table 1 (see Appendix A.1).

	Min	25th	50th	75th	90th	Max	Mean
Corner-to-Corner	0.0	9.0	15.0	24.0	33.0	85.0	17.41
Door-to-Door	0.0	12.0	20.0	29.0	38.0	89.0	21.49

Table 1: Wait Time Distribution of Completed Requests (Jan-Oct 2025)

Additionally, there was a larger frequency of requests made during door-to-door hours. In the same months, there were 112,611 total completed ride requests for door-to-door and only 41,132 total completed ride requests for corner-to-corner. Requests made during corner-to-corner also had a higher completion rate than those made during door-to-door as shown in Table 2.

Service Type	Total Requests	Completed Requests	Completion Rate
Corner-to-Corner	62,711	41,132	65.59%
Door-to-Door	183,890	112,611	61.24%

Table 2: Request Completion Rates (Jan-Oct 2025)

Given the larger proportion of riders affected, higher variability and magnitude of wait times, and lower completion rates between the hours of 8:00PM and 3:00AM, our project focuses specifically on improving wait time performance for door-to-door service. We propose a new dynamic batching and routing strategy that groups incoming ride requests together and assigns these batches to vehicles in rolling time windows. Using historical rider request data, geographic network information, and a pickup-and-delivery Traveling Salesperson formulation, the goal of our proposed solution is to demonstrate how the use of batching and route optimization can lead to significantly reduced wait times without altering vehicle supply count as well as provide insights and recommendations for improving the shuttle’s operations.

3 Data and Preprocessing

Hourly Vehicle Supply Plan: The provided supply plan is only for the month of October and provides the expected number of vehicles out on the road at each hour of operation. Therefore, we only look at the October time frame to run our simulation.

Ride Request Data: Provided by Columbia Transportation and Via, the dataset on ride requests is comprised of pickup and dropoff coordinates, the request arrival time, request creation date, passenger count, and hour of the day. These specific fields are to be used in our model simulation. Looking at completed rides only in the month of October, the distribution of actual wait times (in min) by percentile for the month of October is provided below in Table 3.

Min	25th	50th	75th	90th	Max	Mean
0.0	16.0	26.0	35.0	43.0	79.0	25.94

Table 3: Wait Time Distribution of Door-to-Door Completed Requests in October 2025

Road Network Map Creation (KML + OpenStreetMap): To define the geographical boundaries of the shuttle service area, we used Google MyMaps to create a polygon shape whose lines and vertices represent the outer boundary of the region and exported the coordinate mapping into a KML file (see Appendix A.2). It is also important to note that this polygon extracted from the KML file approximates the actual service boundary. While there is a slight discrepancy, it does not affect our results as all historical ride requests used in the simulation fall within this extracted region. We then download real street network data from OpenStreetMap to identify all drivable streets within the service area specified in the KML file. The resulting graph is a GeoDataFrame comprised of nodes and edges (see Appendix A.3). By doing so, we ensure that the routing and distance calculations used in our model reflect real-world constraints.

4 Model Assumptions

Due to the operational complexity of the shuttle service, assumptions are made regarding the construction of the graph network, batching logic, vehicle assignment rules, and routing optimization to simplify our model.

Ride Requests Each person requesting a ride can add at most 2 passengers. This is in line with Columbia Transportation’s current guidelines for additional passengers (Columbia Transportation, ndb).

Pick up and Drop Off Coordinates: A rider’s pickup coordinates can lie anywhere in the continuous geographic space while vehicles are constrained to a discrete street network. To reconcile these two differences, we map every request’s pick up and drop off coordinates to the closest predefined node on our road network graph that was created using KML and OpenStreetMap (see Appendix A.3). Mapping to the the nearest graph nodes for each pick up or drop off location allows the model to compute the shortest-path distances over the street network. This allows for realistic travel time estimation and vehicle routing.

Vehicle Type: All vehicles are of the same model and size with a maximum seat capacity of 6. We do not take into consideration whether the vehicles are wheelchair accessible.

Vehicle Service Behavior: The model only looks at door-to-door service which starts at 8pm every night and runs until 3am (Columbia Transportation, nda). At 8pm, each vehicle begins at a the depot node which has been predetermined to be at the coordinates (-73.9630, 40.8070). We treat all passenger drop-offs as instantaneous events. Regarding pick ups, drivers wait up to 2 minutes for the passenger to arrive to the pick up spot, and we assume that all passengers make it in time. To model this variability in passenger pick up wait time, we use a random distribution of Uniform(0,2).

Traffic and Travel Times: To represent the changing traffic conditions on the road, vehicle speeds depend on the time of day. These values are deterministic, and we do not use a random

distribution to model the speeds (Appendix B.1). Once the corresponding speed is assigned for a given batch based on the time, all resulting travel times are treated as deterministic and are computed from road network distances. This allows for travel time estimation to remain consistent across optimization runs.

5 Modeling Framework

5.1 Distance and Speed Modeling

We map each request’s pick up and drop off coordinate respectively to the nearest node in our network graph as outlined in Section 3 (Appendix A.3) and then compute the shortest path distance between these two nodes. If no path exists between the two nodes on the graph, a safeguard is in place by alternatively computing the shortest path distance using Euclidean Distance.

5.2 Batching and Sub-Batching

With regards to the batching windows, time is segmented into non-overlapping intervals where each interval is of length Δt minutes. Requests arrive within each time interval, and at the end of each Δt period, the batching window closes. The system freezes all requests that have come through, and this set of requests is what becomes a batch. This batch is then split into sub-batches by iterating through the requests based on order of arrival. The model maintains a running total of passengers in the current sub-batch, and once adding the next request would exceed the vehicle capacity of 6 seats, it closes the current sub-batch and starts a new one (see Appendix B.2). If there are n total requests in a batch, the total number of sub-batches to be created is $\lceil n/6 \rceil$. This batching process prioritizes filling up each vehicle’s load to its maximum capacity.

5.3 Vehicle Assignment

Each sub-batch of requests is assigned to exactly one vehicle. Requests in a sub-batch are never split across multiple vehicles. If the number of sub-batches is less than the number of vehicles, then any vehicle that is not assigned a sub-batch remains idle until needed in future batches. No sub-batch is dropped or merged with other sub-batches.

Once all requests have been assigned to a sub-batch, each sub-batch is then assigned to a vehicle. This assignment looks at the availability of vehicles and follows a greedy heuristic depending on the scenarios highlighted in Appendix B.3. Note that a vehicle is defined as idle when the driver is currently not servicing any requests, while a vehicle is busy when it has been assigned a sub-batch and in the middle of servicing requests. See Appendix B.4 for an example.

5.4 Pickup-and-Delivery TSP Formulation

For every vehicle and assigned sub-batch of requests, we run TSP to determine the best route for each vehicle with an objective of minimizing the total route time traveled. The mathematical formulation detailing all of the parameters, sets, decision variables, objective function, and constraints of the TSP model is in Appendix B.5.

5.5 Successor-Based Route Reconstruction

After Gurobi runs the TSP MIP and produces an optimal solution for a given vehicle and sub-batch of requests, we formulate the route based on the output of the decision variables. The model inspects the values of x_{ij} and reconstructs a path that covers all $2n+1$ pick up and drop off nodes (see Appendix B.6). If reconstruction fails or Gurobi was unable to find an optimal solution, a safeguard is in place where the route is constructed using a simple heuristic where

the vehicle visits all pickups first in order of index followed by all of the drop offs.

5.6 Dispatch of Vehicles

Vehicle routes do not begin before the batching window closes. Even if enough requests arrive to fill a vehicle before Δt minutes, the system does not run sub-batching, vehicle assignments, and TSP in advance. In other words, each idle vehicle must wait until the Δt minutes are up before departing. As for vehicles busy servicing requests from the previous batch, these vehicles cannot leave their current route for their next assignment until they drop off the last request in their route.

6 Model Performance and Validation

To evaluate the robustness of the dynamic batching and routing framework, we perform two sets of sensitivity tests using the historical ride request data and vehicle supply plan for the month of October. Due to time constraints with the size of the model, we only ran the tests using data from the first two weeks of October. We also use all requests in the system including those originally unaccepted or canceled.

6.1 Sensitivity Analysis

6.1.1 Test 1: Effect of Batching Window Size

We test different batching intervals of $\Delta t \in \{2, 5, 8, 10, 15\}$ minutes while keeping all other parameters fixed. Table 4 shows that as Δt increases, our model shows that average predicted ride time increases monotonically from 7.34 to 8.34 minutes. This increase is expected as batching windows generate larger shared routes and expand pooling opportunities. However, across a 13-minute time window difference, the resulting average ride times only differ by around 1 minute which suggests that once Δt is long enough to fill 6 seats in each vehicle, any additional window time cannot produce a longer route. The marginal effect of adding more riders to a route diminishes, producing saturation in predicted ride time growth. In addition, predicted wait time increases as Δt increases. This is expected because with our batching policy, no routes are released until a time window closes. Larger batches also take longer to serve due to more complex routing and vehicle supply constraints. All 10,363 requests were also successfully batched across all Δt , which demonstrates that batching does not affect model feasibility. Our model remains stable and produces consistent routing patterns despite the variation in batching frequency.

Δt	Requests		Ride Time		Wait Time	
	Sim.	Actual	Pred.	Actual	Pred.	Actual
2	10363	10363	7.3449	8.9442	13.6362	24.9977
5	10363	10363	7.8809	8.9442	14.5968	24.9977
8	10363	10363	8.2227	8.9442	16.7858	24.9977
10	10363	10363	8.2787	8.9442	17.6470	24.9977
15	10363	10363	8.3438	8.9442	20.5263	24.9977

Table 4: Sensitivity Test 1 Results

6.1.2 Test 2: Effect of Vehicle Supply Levels

We then test how modifying the hourly vehicle supply plan affects model performance. We consider three scenarios while keeping $\Delta t = 5\text{min}$. The Baseline scenario uses the original supply plan data, while Low Supply is a 20% reduction of the Baseline and High Supply is a 20% increase of the Baseline. All values are rounded down to the nearest integer.

Scenario	Requests		Ride Time		Wait Time	
	Sim.	Actual	Pred.	Actual	Pred.	Actual
Baseline	10363	10363	7.9719	8.9442	14.4876	24.9977
Low Supply	10363	10363	7.9507	8.9442	31.2500	24.9977
High Supply	10363	10363	7.9445	8.9442	12.5474	24.9977

Table 5: Sensitivity Test 2 Results

Under low supply (-20%), average predicted wait time nearly doubles (from 14.4876 to 31.2500 minutes) which implies insufficient fleet coverage. Under high supply ($+20\%$), average predicted wait time decreases minimally from 14.4876 to 12.5474 minutes. Furthermore, predicted ride times remain relatively stable with minimal differences across the three scenarios. This is because vehicle availability affects when a passenger starts riding and not how long the route takes once it begins. The small amount of noise observed comes from batching randomness, sequencing variability, and the Uniform(0,2) passenger delay. All 10,363 requests in the simulated time frame from October 1st to 14th were also serviced. Fleet supply is therefore sufficient across this horizon, so batching does not introduce any unmet demand. None of the vehicles were overloaded at any point in the simulation. Please refer to Appendix C.1 and C.2 for a deeper analysis on the predicted ride time and wait time distributions for both sensitivity tests.

6.2 Confidence Interval Evaluation

We computed the 95% confidence intervals for each Δt and supply level for predicted ride time and predicted wait time to better understand the results of our sensitivity tests. Because the stochastic variability introduced in the model is minimal (i.e. applying a Uniform(0,2) distribution to driver pick up wait times), we ultimately determined that the individual CIs are sufficient for inference. In empirical settings with higher noise levels, we would compute CIs on the mean differences.

Table 6 shows that for predicted wait times, none of the confidence intervals overlap as Δt changes. Increasing Δt leads to meaningful increases in expected wait time, which suggest that these changes are large relative to simulation noise. Hence, the value of the batching window is a primary lever in controlling wait times.

Additionally, the non-overlapping intervals for predicted ride time when Δt is small enough at 2 or 5 mean that lower batching frequencies can improve routing enough to reduce ride times by some amount. With larger batching windows, however, this may not necessarily be the case as these respective CIs for predicted time do overlap. Routing structure and geographical distances govern ride time instead of batching frequency as any further increase in batching window does not change the routes substantially enough.

Δt	Predicted Ride Time	Predicted Wait Time
2	[7.19, 7.40]	[12.74, 13.04]
5	[7.83, 8.04]	[14.15, 14.45]
8	[8.10, 8.32]	[16.19, 16.50]
10	[8.18, 8.39]	[17.07, 17.38]
15	[8.25, 8.47]	[19.98, 20.30]

Table 6: 95% Confidence Intervals by Δt

For our vehicle supply plan sensitivity analysis results, all three confidence intervals for

predicted ride time are extremely narrow and overlap almost perfectly. Routing remains efficient regardless of vehicle availability as increasing supply does not shorten any of the travel times. For predicted wait time, there are no overlapping CIs meaning there are statistically meaningful differences in wait times across the different supply levels.

Vehicle Supply	Predicted Ride Time	Predicted Wait Time
Baseline	[7.87, 8.08]	[14.34, 14.64]
Low Supply (-20%)	[7.85, 8.06]	[30.92, 31.58]
High Supply (+20%)	[7.84, 8.05]	[12.43, 12.67]

Table 7: 95% Confidence Intervals by Vehicle Supply Level

6.3 Model Stability Under Temporal Demand Variation

To further test robustness of our model, we performed additional sensitivity tests on data covering October 15th to October 31st. The goal of running these additional tests is to compare the model’s performance between the first 2 weeks and the last 2 weeks of the month. Results show that the system remains stable with respect to temporal variation in demand (see Appendix C.4 for details).

7 Discussion

Across all batching scenarios tested, predicted wait times were consistently lower than the actual wait times recorded in the ride request data. Although batching holds requests until the window closes, our simulation shows that despite this small delay, average wait time remains below historical levels which indicates that routing efficiency outweighs batching-induced delay. Based on our simulation results, adopting a batching window of 5 minutes without requiring any increase in fleet size could reduce average wait times by roughly 43% (10.7 minute decrease) relative to historical door-to-door performance (see Appendix C.3).

We also acknowledge that our model is not without its limitations due to project time constraints and modeling assumptions made. Our model can be improved further by optimizing the sub-batch splitting process described in Section 5.2. We would recommend employing a clustering method where pick up and drop off nodes in close proximity to each other are more likely to be assigned to the same vehicle. In addition, our vehicle assignment process covered in Section 5.3 focuses on maximizing each vehicle’s passenger load which may not be the most efficient approach. For example, if only 5 requests have arrived in a given batching time window, our model assigns all 5 requests to 1 vehicle instead of distributing them to multiple idle vehicles. By considering a load-balancing assignment rule, our model could further reduce average wait times.

8 Conclusion

By coordinating requests within decision horizons, Columbia Transportation can reduce average wait times while simultaneously serving all demand and reinforcing its safety mission. These improvements are achieved with more deliberate routing and batching policies without the need to expand existing fleet. Based on the model’s results, we recommend adopting a batching horizon between 5 to 8 minutes as it balances pooling efficiency with passenger wait time delays. Our approach represents a practical and a cost-efficient policy tool that Columbia Transportation can adjust according to their needs to further boost ridership and service quality.

References

- Columbia Transportation (n.d.a). Evening Shuttle (Via) Service Monthly Start Times. <https://transportation.columbia.edu/content/evening-shuttle-service-monthly-start-times>. Accessed Nov 24 2025.
- Columbia Transportation (n.d.b). On-Demand Evening Shuttle Frequently Asked Questions (FAQs). <https://transportation.columbia.edu/content/demand-evening-shuttle-frequently-asked-questions-faqs>. Accessed Nov 23 2025.

Appendix

A Data Analysis and Preprocessing

A.1 Ride Wait Time Distributions

From the two graphs below, it can be observed that both distributions are right-skewed with tails. Most riders experience moderate wait times, but there is a small percentage of those who experience pretty long waits that range from 40 to up to 80 minutes. Currently, there is significant variability in wait times, and performance based on the ride request data suggests that the system may struggle either during peak demand hours or when vehicle supply is limited.

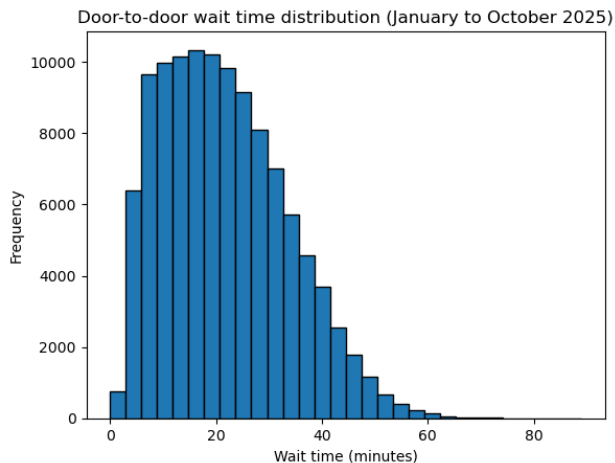


Figure 1: Door-to-Door Wait Time Distribution (January to October 2025)

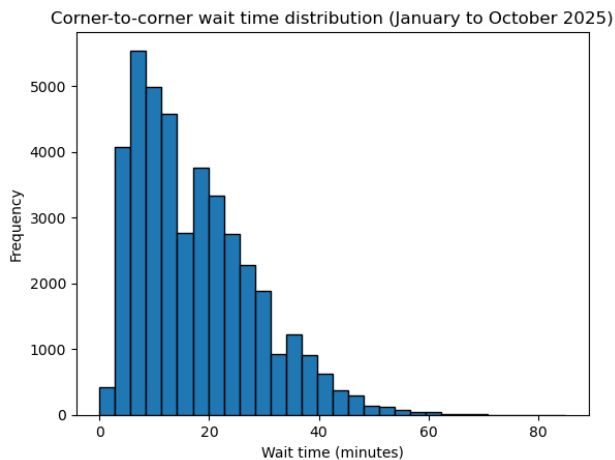


Figure 2: Corner-to-Corner Wait Time Distribution (January to October 2025)

A.2 KML Polygon

Below is a screenshot of the KML polygon that the team has created in Google MyMaps. We then use this polygon exported in the form of a KML file to later build our road network map illustrated in Appendix A.3 - Figure 4.

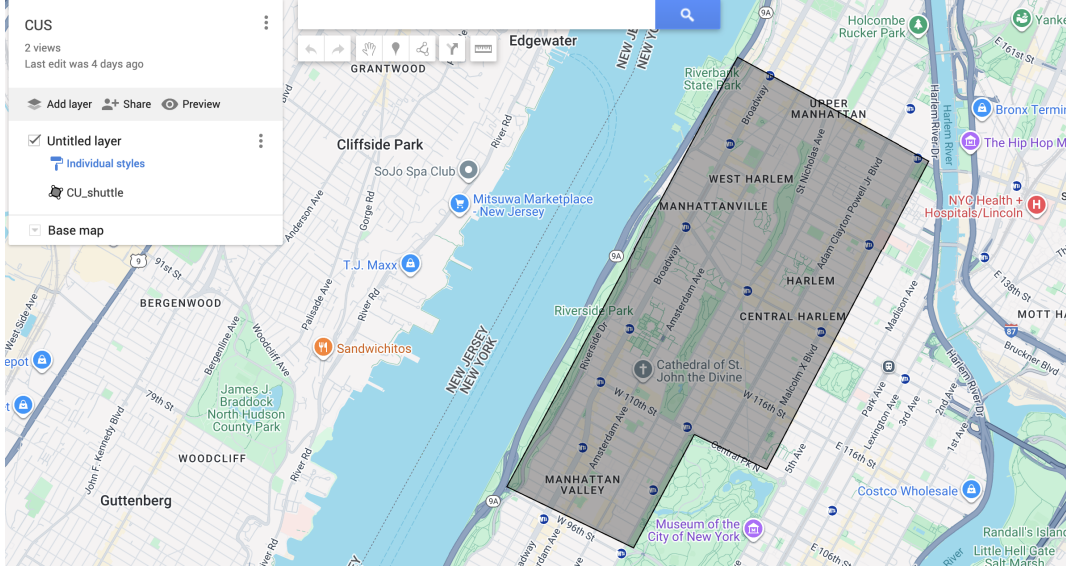


Figure 3: Shuttle service area as defined by the KML polygon.

A.3 OpenStreetMap Graph Construction

We create the below graph using the OpenStreetMap package and KML polygon shape created in Appendix A.3. This graph represents the coverage area of Columbia's night shuttle service, and it contains predefined nodes mapped along the streets which represents possible pickup and dropoff nodes. Because a rider can enter any pick up or drop off coordinate possible (including buildings), we create these predefined nodes to standardize all possible pick up and drop off nodes to make it easier to compute the distances between nodes, total travel time, and more. Our graph is comprised of 1645 nodes and 2006 edges in total.

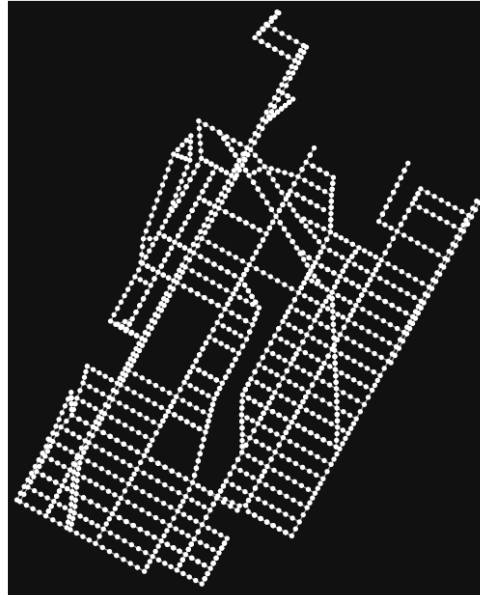


Figure 4: Network Graph with Predefined Nodes

B Modeling Framework

B.1 Distance and Speed Modeling

To take into account traffic, driving speed (km/h) is selected based on the average hour of day of the requests in a given batch. Let average hour of day equal `hour_of_day`. Then we specify the deterministic speeds as:

- If $4\text{pm} \leq \text{hour_of_day} \leq 7\text{pm}$, vehicle speed is 20 km/h.
- If $7\text{pm} \leq \text{hour_of_day} \leq 9\text{pm}$, vehicle speed is 18 km/h.
- If $9\text{pm} \leq \text{hour_of_day} \leq 11\text{pm}$, vehicle speed is 22 km/h.
- If $11\text{pm} \leq \text{hour_of_day} \leq 3\text{am}$, vehicle speed is 28 km/h.

B.2 Batching and Sub-Batching Example

To simulate the batching and sub-batching process, consider the requests made on October 1, 2025 from `Ride Requests_2025-11-11-part-2.csv`. The first time window where $\Delta t = 5$ minutes is right when service starts from 8:00 PM to 8:04 PM. In total, there are 22 requests made in this time frame which together form 1 specific batch. Let these 22 requests be defined as Request 0, 1, ..., 21 where Request 0 is the first to arrive to the system and Request 21 is the last before the batching window closes. Requests 7 and 21 have 2 passengers, while all other requests only have 1 passenger.

This batch is then split into sub-batches to later be assigned to the vehicles. Because the sub-batching process is based on the order of request arrivals and the 6-seat maximum vehicle capacity, there are $\lceil 22/6 \rceil = 4$ sub-batches created. Then let the sub-batches be defined with the index 0, 1, ..., 3. The batch is then split into 4 sub-batches as shown in Table 6. Once 6 passengers* fill up a sub-batch, a new sub-batch is created. This process is repeated until all requests in the batch have been assigned to a sub-batch.

*Note that this is 6 passengers and not 6 requests since there can be more than 1 passenger per request.

Sub-Batch	Requests	Passenger Count
0	0, 1, 2, 3, 4, 5	6
1	6, 7, 8, 9, 10	6
2	11, 12, 13, 14, 15, 16	6
3	17, 18, 19, 20, 21	6

Table 8: Assignment of Requests to Sub-Batches

B.3 Vehicle Assignment Scenarios

The scenarios below describe in more detail how the batching and sub-batching processes work depending on the availability of vehicles:

Scenario 1 - All vehicles are idle which occurs during the very first time window when service has just begun: If the number of sub-batches is less than or equal to the number of vehicles, assign by order of index each sub-batch to each vehicle. For example, sub-batch 0 is assigned to vehicle 0, sub-batch 1 is assigned to vehicle 1, and so forth. If the number of sub-batches exceeds the number of vehicles, each available vehicle is assigned a sub-batch, and any remaining unassigned sub-batches must wait until the next batching period and when a vehicle becomes available.

Scenario 2 - There is a combination of idle vehicles and busy vehicles: Each idle vehicle is assigned a sub-batch first before each busy vehicle. The assignment of idle vehicles is the same as the first scenario which is based on the order of index. Among the pool of busy vehicles, the model assigns a sub-batch to the vehicle that offers the earliest possible pick up time. For each busy vehicle, the model computes the distance between the vehicle's last drop-off location in its current route and every pick up node in the new sub-batch of requests. The pick up node with the shortest distance is then selected in the computation of the earliest possible pick up time for that vehicle. Then, across all busy vehicles, the vehicle with the shortest pick up time is selected and assigned that sub-batch of requests to service after it completes its current sub-batch.

Scenario 3 - All vehicles are busy: The assignment of sub-batch to a busy vehicle is the same as described in Scenario 2.

B.4 Vehicle Assignment Example

Continuing with the example from Appendix B.2, we now look at how our model assigns each sub-batch to a vehicle.

The supply plan in `CUL October 2025 Supply Plan.xlsx` shows that there are 10 vehicles out servicing requests between 8pm to 9pm on October 1st, and they are defined in the model as Vehicles 0, 1, ..., 9.

Because the vehicles have just started service, they have not yet been assigned any requests and are therefore idle during this initial batching. The assignment process therefore follows Scenario 1 outlined in Section 5.3 where we start by assigning sub-batch 0 to Vehicle 0. Because the number of sub-batches is less than the the number of vehicles, we simply assign the sub-batches as follows:

- Sub-batch 0 $\rightarrow$ Vehicle 0
- Sub-batch 1 $\rightarrow$ Vehicle 1
- Sub-batch 2 $\rightarrow$ Vehicle 2
- Sub-batch 3 $\rightarrow$ Vehicle 3

The remaining 5 vehicles (Vehicles 5, 6, ..., 9) stay idle until the next round of batching.

B.5 Pickup and Delivery TSP Model Formulation

The TSP model for each vehicle and assigned sub-batch is defined as follows:

Sets and Indices

- n = number of requests
- $N = 2n + 1$ = total number of nodes
- Node 0: depot (starting location) for the vehicle
- Nodes 1, 2, ..., n : pick up nodes
- Nodes $n + 1, n + 2, \dots, 2n$: drop off nodes
- For request $r \in \{0, 1, \dots, n - 1\}$:
 - Pickup node: $p_r = 1 + r$
 - Dropoff node: $d_r = 1 + n + r$
- G = road network graph created as shown in Figure 4
 - V = set of all vertices (nodes) in graph G (representing the pick up and drop off nodes)
 - E = set of all edges in graph G (representing the streets)

Parameters

- c_{ij} : travel time from node i to node j (minutes)
- q_j : change in vehicle load at node j where
 - $q_0 = 0$ (depot)
 - $q_j = +\text{num_passengers}_r$ for $j = p_r$ (pickup)
 - $q_j = -\text{num_passengers}_r$ for $j = d_r$ (dropoff)
- $0 \leq Q \leq 6$: vehicle capacity
- $t_0 = \text{start time}$
- $M_{\text{time}} = \text{big-M}$ for time constraints (10^6)
- $M_{\text{load}} = \text{big-M}$ for load constraints ($K + 1$)
- $M_{\text{mtz}} = \text{big-M}$ for MTZ constraints ($2n$)

Decision Variables

- $x_{ij} = \begin{cases} 1 & \text{if the vehicle travels directly from node } i \text{ to node } j, \\ 0 & \text{otherwise,} \end{cases} \quad \forall (i, j) \in E$
- L_i : vehicle load after visiting node i

$$L_i \in [0, Q] \quad \forall i \in \{0, 1, \dots, 2n\} \quad (1)$$

- u_i : position of node i in the tour (MTZ ordering variable)

$$u_i \in [0, 2n] \quad \forall i \in \{0, 1, \dots, 2n\} \quad (2)$$

- T_i : time of arrival at node i

$$T_i \geq 0 \quad \forall i \in \{0, 1, \dots, 2n\} \quad (3)$$

Objective Function

$$\min \sum_{i=0}^{2n} \sum_{j=0}^{2n} c_{ij} \cdot x_{ij} \quad (4)$$

Constraints

Flow Conservation (Route Structure)

Each non-depot node has exactly one incoming arc:

$$\sum_{\substack{i=0 \\ i \neq j}}^{2n} x_{ij} = 1 \quad \forall j \in \{1, 2, \dots, 2n\} \quad (5)$$

Each non-depot node has at most one outgoing arc:

$$\sum_{\substack{j=0 \\ j \neq i}}^{2n} x_{ij} \leq 1 \quad \forall i \in \{1, 2, \dots, 2n\} \quad (6)$$

One arc must leave the depot:

$$\sum_{j=1}^{2n} x_{0j} = 1 \quad (7)$$

No arc returns to the depot:

$$\sum_{j=0}^{2n} x_{j0} = 0 \quad (8)$$

No self-loops:

$$x_{ii} = 0 \quad \forall i \in \{0, 1, \dots, 2n\} \quad (9)$$

MTZ-Subtour Elimination

Initialize the depot node to be the first stop in the tour:

$$u_0 = 0 \quad (10)$$

Ordering variable bounds:

$$1 \leq u_i \leq 2n \quad \forall i \in \{1, 2, \dots, 2n\} \quad (11)$$

If $x_{ij} = 1$, then $u_j \geq u_i + 1$, j must come after i in the tour:

$$u_j \geq u_i + 1 - M_{\text{mtz}}(1 - x_{ij}) \quad \forall i, j \in \{1, 2, \dots, 2n\}, i \neq j \quad (12)$$

Time Propagation

Initial time:

$$T_0 = t_0 \quad (13)$$

Time continuity: If $x_{ij} = 1$, then $T_j \geq T_i + c_{ij}$

$$T_j \geq T_i + c_{ij} - M_{\text{time}}(1 - x_{ij}) \quad \forall i, j \in \{0, 1, \dots, 2n\}, i \neq j \quad (14)$$

Load Propagation

Initial load:

$$L_0 = 0 \quad (15)$$

Load continuity: If $x_{ij} = 1$, then $L_j \geq L_i + q_j$

$$L_j \geq L_i + q_j - M_{\text{load}}(1 - x_{ij}) \quad \forall i, j \in \{0, 1, \dots, 2n\}, i \neq j \quad (16)$$

Capacity Bounds

$$L_i \leq K \quad \forall i \in \{0, 1, \dots, 2n\} \quad (17)$$

$$L_i \geq 0 \quad \forall i \in \{0, 1, \dots, 2n\} \quad (18)$$

Pickup-Before-Dropoff Precedence

For each request r , the dropoff node must come after the pickup node in the ordering of the tour.

$$u_{d_r} \geq u_{p_r} + 1 \quad \forall r \in \{0, 1, \dots, n-1\} \quad (19)$$

B.6 Successor-Based Route Reconstruction

The TSP model is run to determine the best route the minimizes total time traveled for a given vehicle. Taking the results from this model, we use the decision variable outputs to construct the route that the vehicle takes. To do so, we continue with the example from Appendix B.2 and B.4.

Previously, vehicle 0 was assigned sub-batch 0 with requests 0, 1, 2, 3, 4, and 5. The output from running TSP on this vehicle is: [0, 2, 6, 1, 4, 10, 5, 7, 12, 8, 3, 9, 11].

Provided the below information on what each node represents, vehicle 0’s route is as follows: pick up request 1, pick up request 5, pick up request 0, pick up request 3, drop off request 3, pick up request 4, drop off request 0, drop off request 5, drop off request 1, pick up request 2, drop off request 2, and then finally drop off request 4.

- Node 0: Depot
- Node 1: Request 0’s pick-up node
- Node 2: Request 1’s pick-up node
- Node 3: Request 2’s pick-up node
- Node 4: Request 3’s pick-up node
- Node 5: Request 4’s pick-up node
- Node 6: Request 5’s pick-up node
- Node 7: Request 0’s drop-off node
- Node 8: Request 1’s drop-off node
- Node 9: Request 2’s drop-off node
- Node 10: Request 3’s drop-off node
- Node 11: Request 4’s drop-off node
- Node 12: Request 5’s drop-off node

C Model Performance and Validation

C.1 Sensitivity Test 1: Batch Time Windows

The sensitivity tests for varying batch time windows were conducted using ride request data from October 1st to 14th. Below are the distributions for predicted ride time and wait times.

Δt	Min	25th	50th	75th	90th	Max	Mean
2	0	3.316601	5.739616	9.892098	14.820057	32.551437	7.294759
5	0	3.769314	6.493381	10.883175	15.593051	36.625935	7.931370
8	0	3.903978	6.751697	11.272994	16.403791	39.043211	8.209122
10	0	3.942507	6.846375	11.416598	16.222578	36.086616	8.285672
15	0	3.976044	7.019605	11.521031	16.439108	35.426804	8.358452

Table 9: Predicted Ride Time Distribution for Varying Δt

Note that the observed minimum predicted ride of 0 minutes is not a design flaw in the model. This occurs when a request’s pickup and dropoff nodes are either adjacent in the optimized route

or extremely close together geographically, which results in a rounded down in-vehicle time of 0 minutes. These are not common in the simulation and only reflect edge cases where the travel time is small enough to where it has been rounded down to zero. This reinforces that most variability in performance comes from wait time and not travel time where ride times are constrained by the network itself than by batching or vehicle assignment.

Δt	Min	25th	50th	75th	90th	Max	Mean
2	1.109062	6.536366	11.024790	17.659666	24.574572	51.043431	12.889904
5	1.032606	8.405640	12.807591	18.778673	24.878716	50.173401	14.301185
8	1.219132	10.280073	14.955268	21.280743	27.352005	55.991066	16.3440159
10	1.627849	11.461721	16.037491	22.126877	28.217540	51.349206	17.223503
15	1.041809	14.080669	19.296804	25.606627	31.606790	53.551600	20.140346

Table 10: Predicted Wait Time Distribution for Varying Δt

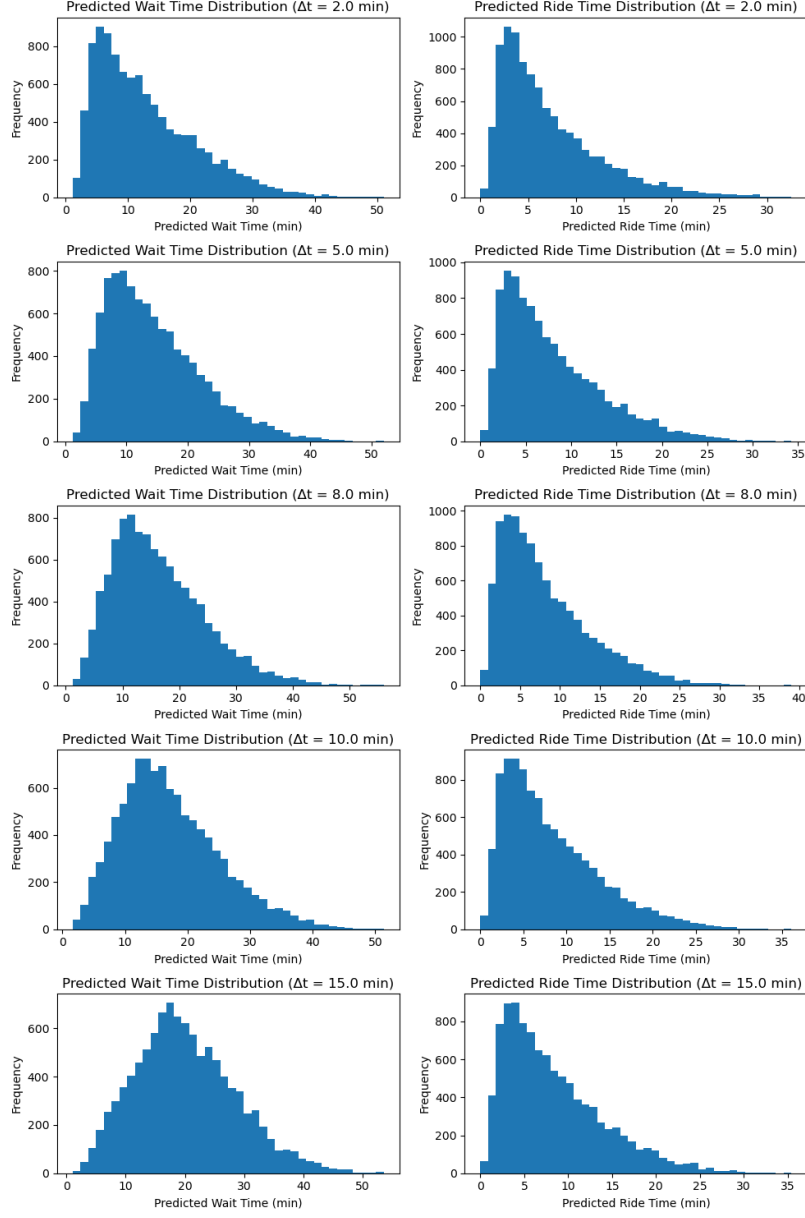


Figure 5: Predicted Wait and Ride Time Distribution Plots from Δt Sensitivity Analysis

Overall, all distributions appear unimodal which suggests that the system is stable across Δt . Looking at the predicted wait time distribution plots on the left hand side of Figure 5, it can be seen that as Δt increases, the wait time distributions shift to the right as seen with the peak (mode) of each distribution. With regards to the changes in shape, at the smallest Δt value of 2 minutes, there is a larger skew with a more prominent right tail. This means that a smaller subset of riders waited an unusually long amount of time. More riders experience moderate delays when batching windows are bigger.

The predicted ride time distributions on the right hand side change minimally across Δt as discussed in the report. The mean moves only slightly, and the overall shape stays the same. The tail length is also stable and does not fluctuate drastically across different batching time windows.

C.2 Sensitivity Test 2: Vehicle Supply Plan Levels

The sensitivity tests for varying batch time windows were also conducted using ride request data from October 1st to 14th. Below are the distributions for predicted ride time and wait times.

Vehicle Supply Level	Min	25th	50th	75th	90th	Max	Mean
Baseline	0	3.704611	6.515509	10.993934	15.871270	34.263807	7.971996
Low Supply (-20%)	0	3.754220	6.549739	10.857679	15.717736	36.208133	7.950689
High Supply (+20%)	0	3.760529	6.462176	10.889410	15.711885	36.578621	7.944516

Table 11: Predicted Ride Time Distribution for Varying Vehicle Supply Level

Vehicle Supply Level	Min	25th	50th	75th	90th	Max	Mean
Baseline	1.204497	8.435256	12.947295	19.126307	25.366550	52.008312	14.487573
Low Supply (-20%)	1.466527	17.618675	29.829830	42.183800	55.837262	93.529705	31.250042
High Supply (+20%)	1.100635	7.698901	11.423154	16.395313	21.411016	42.611840	12.547415

Table 12: Predicted Wait Time Distribution for Varying Vehicle Supply Level

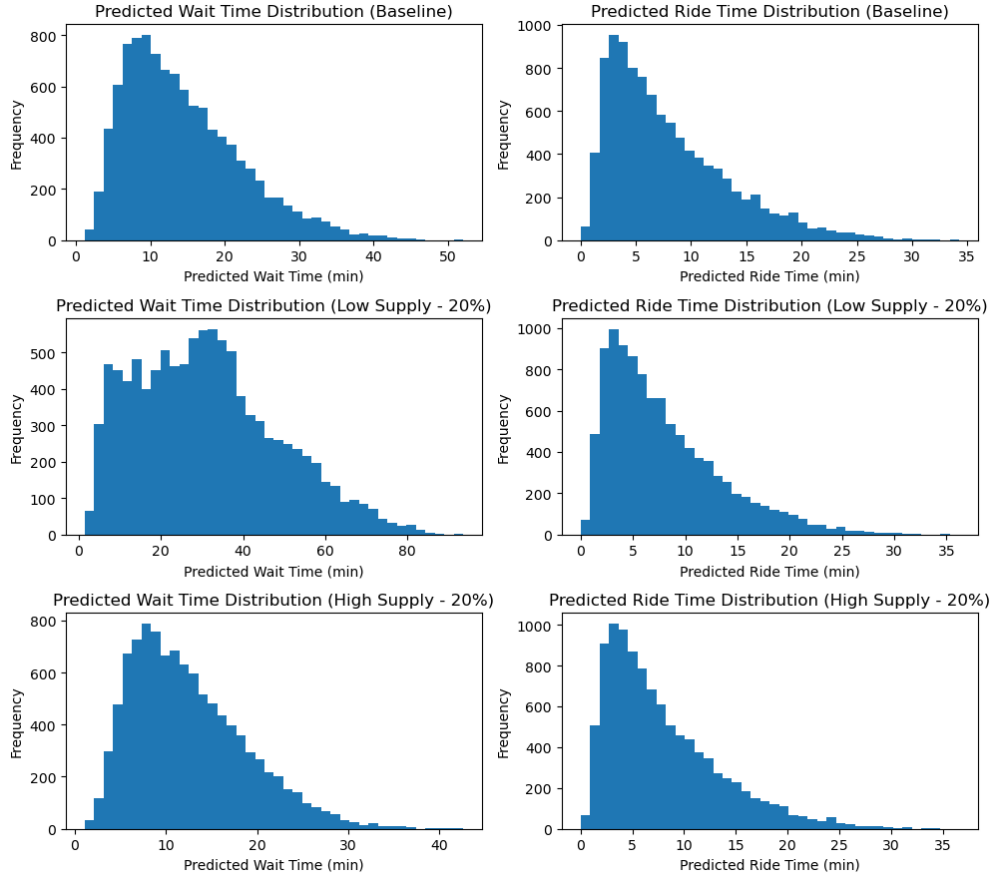


Figure 6: Predicted Wait and Ride Time Distribution Plots by Vehicle Supply Plan Level

For the baseline supply level, the predicted wait time distribution is right-skewed and unimodal with its peak visibly sitting at around 10 minutes. This means that the model meets demand without severe queuing. There are however a few outliers that produce long waits, but based on the 75th percentile, most riders appear to only wait up to 20 minutes. The predicted ride time distribution similarly is right-skewed and appears to be concentrated at around 5 or so minutes.

For the low supply level, the distribution in predicted wait time becomes flattened and irregular, and the peak shifts closer towards 30-35 minutes. These changes show that by reducing the vehicle supply, the dispatch system starts to break down with wait times becoming increasingly unpredictable. As for the predicted ride time distribution at low supply, the shape of the distribution changes minimally with the right skew preserved. This again reinforces the idea that the system struggles with wait times before pick up and not during the actual ride trips themselves.

For the high supply level, the predicted wait time distribution shows that the peak closely at around 10 minutes. Variance also shrinks with a visibly tighter distribution, and the tail is not as prominent. Looking at the scale of the plot, the tail extends only up to 43 minutes whereas the maximum wait time from baseline was 52 minutes. This means that there are significantly fewer long waits across the board. For predicted ride time distribution, it again remains unchanged like in the other scenarios.

All of these results combined show that wait time is sensitive to supply availability and confirms that, with our model, batching is stable under normal or high vehicle supply which is to be expected. As discussed previously in the report, these results also confirm that ride time is governed by geography and routing efficiency, and not supply level.

C.3 Average Wait Time Calculation between Historical and Simulated

A comparison of the wait time distributions between the historical data and the simulated data is provided in the table below. Please note that this is for a batching time window of 5 minutes and because the simulation ran between October 1st and 14th, the historical numbers were also computed using this same time frame. For the first weeks in October, the average wait time therefore improves from approximately 25.0 minutes to 14.30 minutes. This is a 42.8% reduction (10.7 minutes).

	Min	25th	50th	75th	90th	Max	Mean
Historical	1.00	16.00	25.00	34.00	42.00	69.00	24.9977
Simulated	1.032606	8.405640	12.807591	18.778673	24.878716	50.173401	14.301185

Table 13: Wait Time Comparison between Historical vs Simulated (Door-to-Door)

C.4 Additional Sensitivity Analyses on October 15-31 Data

Previously, we only ran sensitivity tests using data from October 1st to October 14th. To further test robustness of our model, we performed additional sensitivity tests on data covering October 15th to October 31st. The goal of running these additional tests is to compare the results between the first 2 weeks and the last 2 weeks. The results below show how our model performs with respect to potential temporal variation in demand.

C.4.1 Analysis by Vehicle Supply Plan Level

Scenario	Requests		Ride Time		Wait Time	
	Sim.	Actual	Pred.	Actual	Pred.	Actual
Baseline	13314	13314	8.090877	9.1925	15.857013	26.6621
Low Supply	13314	13314	8.051539	9.1925	39.676828	26.6621
High Supply	13314	13314	8.081778	9.1925	12.840727	26.6621

Table 14: Average Ride and Wait Times by Supply Level on Oct 15-31 Data

Vehicle Supply Level	Min	25th	50th	75th	90th	Max	Mean
Baseline	0	3.846750	6.660731	11.100147	16.104319	35.419215	8.090877
Low Supply (-20%)	0	3.785214	6.575588	11.039619	16.050397	36.190994	8.051539
High Supply (+20%)	0	3.809962	6.662631	11.060813	16.144920	36.713834	8.081778

Table 15: Predicted Ride Time Distribution for Vehicle Supply Level on Oct 15-31 Data

Vehicle Supply Level	Min	25th	50th	75th	90th	Max	Mean
Baseline	1.007827	9.142329	14.052862	20.706053	27.930213	59.298843	15.857013
Low Supply (-20%)	1.895752	21.749325	36.967224	51.866874	71.669276	138.068832	39.676828
High Supply (+20%)	1.030731	7.894542	11.618852	16.671398	21.891685	44.204926	12.840727

Table 16: Predicted Wait Time Distribution for Vehicle Supply Level on Oct 15-31 Data

C.4.2 Analysis by Δt

Δt	Requests		Ride Time		Wait Time	
	Sim.	Actual	Pred.	Actual	Pred.	Actual
2	13314	13314	7.409179	9.1925	14.829758	26.6621
5	13314	13314	8.056782	9.1925	15.900725	26.6621
8	13314	13314	8.213147	9.1925	17.206334	26.6621
10	13314	13314	8.362267	9.1925	18.531166	26.6621
15	13314	13314	8.405384	9.1925	21.667038	26.6621

Table 17: Average Ride and Wait Times by Δt on Oct 15-31 Data

Δt	Min	25th	50th	75th	90th	Max	Mean
2	0	3.419433	5.891867	10.036682	15.131605	36.847997	7.409179
5	0	3.826555	6.608786	11.102648	15.971464	36.170867	8.056782
8	0	3.904952	6.876196	11.417539	16.077662	34.804324	8.213147
10	0	3.976363	7.029718	11.520576	16.368997	36.119386	8.362267
15	0	4.018481	7.041529	11.705426	16.330115	34.988252	8.405383

Table 18: Predicted Ride Time Distribution for Δt on Oct 15-31 Data

Δt	Min	25th	50th	75th	90th	Max	Mean
2	1.105290	7.352435	12.667344	20.287884	27.800848	59.579352	14.829758
5	1.229311	9.086813	14.112937	20.680267	27.957085	60.131478	15.900724
8	1.573803	10.703733	15.779339	22.138532	28.994387	59.161724	17.206333
10	1.944106	12.150116	17.199435	23.671797	30.341556	60.642987	18.531166
15	1.406476	15.000957	20.539035	27.350676	34.279412	62.245218	21.667038

Table 19: Predicted Wait Time Distribution for Δt on Oct 15-31 Data

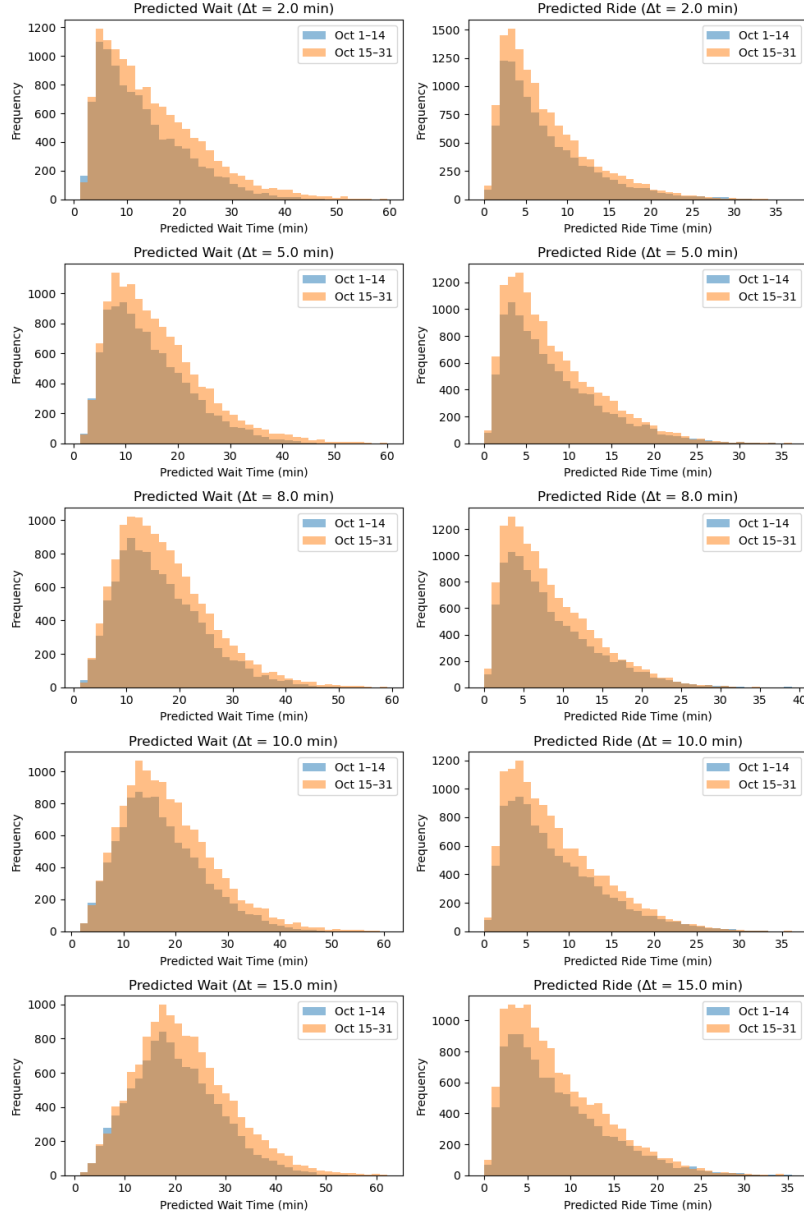


Figure 7: October Comparison of Predicted Time Distributions for Δt

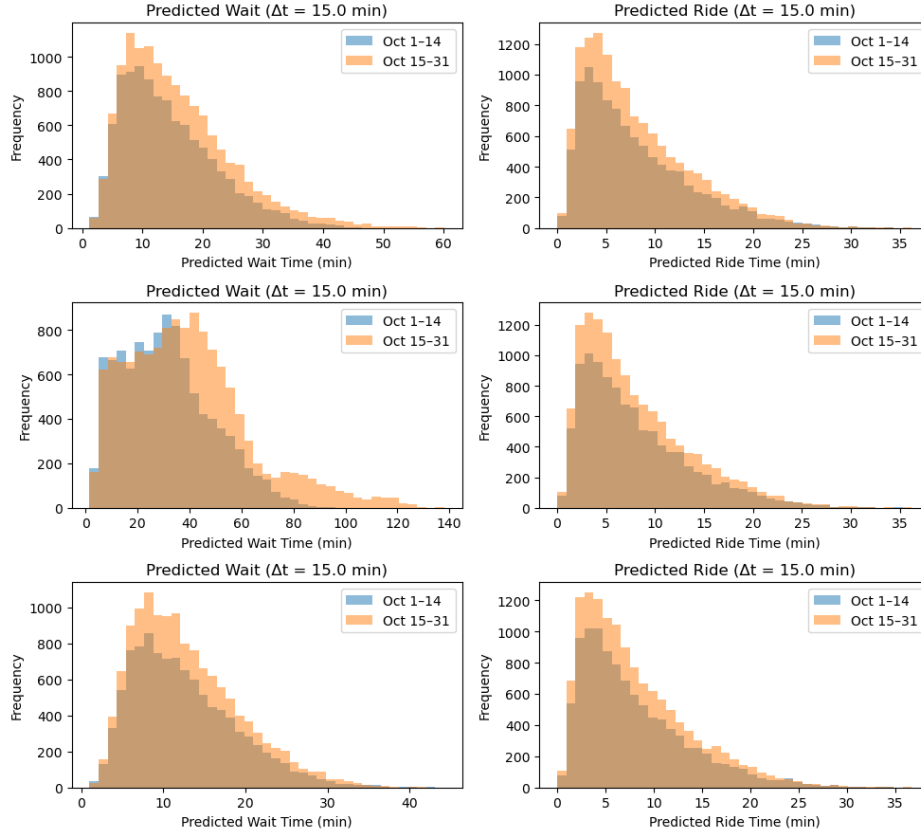


Figure 8: October Comparison of Predicted Time Distributions for Supply Level

There were 13,314 requests in the last 2 weeks of October compared to 10,363 from the first 2 weeks which shows that there was an increase in demand, likely due to the changes in weather and sunset hours. As shown in Figure 7 and Figure 8, the model generalizes across time in the month of October where predicted ride and wait time distributions remain stable despite the demand fluctuation.